

## TOPIC: STATISTICS I

### Meaning and computation of mean of ungrouped data

The mean, median and the mode are called measures of central tendency or measures of location. The mean is also known as the average, the median is the middle number while the mode is the most frequent element or data.

#### THE ARITHMETIC MEAN:

The arithmetic mean is the sum of the ungroup of items divided by the number of it. The mean of an ungrouped data can be calculated by using the formula;

$$\bar{x} = \frac{\sum x}{n}, \quad (\text{when } n \text{ is small}) \quad (\text{where the symbol } \sum x \text{ is called sigma meaning summation of all the given data})$$

$$\text{Also, Mean, } \bar{x} = \frac{\sum k}{\sum f} \quad (\text{when } n \text{ is large})$$

$\sum fx$  = Sum of the product of scores and their corresponding frequencies

$\sum f$  = Sum of the frequencies

Example 1:

Find the arithmetic mean of the numbers 42, 50, 59, 38, 41, 86 and 56

Solution: Add all the numbers and divide by 7

$$\begin{aligned}\bar{x} &= \frac{42 + 50 + 59 + 38 + 41 + 86 + 56}{7} \\ &= \frac{378}{7} \\ \therefore \bar{x} &= 54\end{aligned}$$

Example 2:

The table below gives the frequency distribution of marks obtained by some students in a scholarship examination.

|           |    |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|----|
| Scores(x) | 15 | 25 | 35 | 45 | 55 | 65 | 75 |
| Frequency | 1  | 4  | 12 | 24 | 18 | 8  | 3  |

Calculate, correct to 3 significant figures the mean mark of the distribution (WAEC)

**Solution:**

| Scores(x)     | Frequency |                  |
|---------------|-----------|------------------|
| 15            | 1         | 15               |
| 25            | 4         | 100              |
| 35            | 12        | 420              |
| 45            | 24        | 1080             |
| 55            | 18        | 990              |
| 65            | 8         | 520              |
| 75            | 3         | 225              |
| $\sum f = 70$ |           | $\sum fx = 3350$ |

$$\text{Since Mean; } \bar{x} = \frac{\sum fx}{\sum f}$$

$$\begin{aligned}\bar{x} &= \frac{3350}{70} \\ \bar{x} &= 47.857143\end{aligned}$$

$$\bar{x} = 47.9 \quad (3\text{s.f})$$

$$\begin{aligned}\text{Method 2: mean; } \bar{x} &= \frac{\sum fx}{\sum f} = \frac{(1 \times 15) + (4 \times 25) + (12 \times 35) + (24 \times 45) + (18 \times 55) + (8 \times 65) + (3 \times 75)}{1 + 4 + 12 + 24 + 18 + 8 + 3} \\ &= \frac{15 + 100 + 420 + 1080 + 990 + 520 + 225}{70} \\ &= 47.8571\end{aligned}$$

$$= 47.9 \quad (3\text{s.f})$$

### Meaning and computation of median of ungrouped data

The median is the value of the middle item when the items are arranged in order of magnitude either ascending or descending order.

Example 1;

Find the median of the following set of numbers; 16, 13, 10, 23, 36, 9, 8, 48, 24

Solution: Arrange in (either ascending or descending order)

8, 9, 10, 13, 16, 23, 24, 36, 48

The middle number is 16

$\therefore \text{median} = 16$

Median from frequency distribution (i.e when N is large)

Median =  $\left(\frac{N+1}{2}\right)^{\text{th}}$ , when N is odd

Median =  $\frac{\left(\frac{N}{2}\right)^{\text{th}} + \left(\frac{N}{2} + 1\right)^{\text{th}}}{2}$ , when N is even

Example 2:

The table below shows the distribution of marked scored by some students in a maths test

|           |    |    |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|----|----|
| Marks %   | 22 | 24 | 36 | 42 | 45 | 48 | 56 | 60 |
| Frequency | 11 | 2  | 7  | 13 | 10 | 3  | 9  | 5  |

Solution:

To find the median, a cumulative frequency table is needed.

| Marks %(x) | Frequency | Cumulative frequency |
|------------|-----------|----------------------|
| 22         | 11        | 11                   |
| 24         | 2         | 13                   |
| 36         | 7         | 20                   |
| 42         | 13        | 33                   |
| 45         | 10        | 43                   |
| 48         | 3         | 46                   |
| 56         | 9         | 55                   |
| 60         | 5         | 60                   |

From the table, there are 60 members as indicated by the cumulative frequency.

$$\begin{aligned}
 \text{Since 60 is even, Median} &= \frac{(\frac{N}{2})th + (\frac{N}{2} + 1)th}{2} \\
 &= \frac{(\frac{60}{2})th + (\frac{60}{2} + 1)th}{2} \\
 &= \frac{30th + 31st}{2}
 \end{aligned}$$

The 30<sup>th</sup> member is 42% and the 31<sup>st</sup> member is 42%

$$\therefore \text{median} = \frac{42\% + 42\%}{2} = \frac{84}{2} = 42\%$$

### Meaning and computation of mode of ungrouped data

The mode of a given data is the item which occurs most often in the distribution

Example 1;

The record of the marks scored by a number of students in an oral test in economics is as follows; 10, 10, 5, 9, 15, 10, 20, 10, 9, 5, 9, 10, 25, 9, 5, 25. Find the modal mark

Solution:

|           |   |   |    |    |    |    |
|-----------|---|---|----|----|----|----|
| Marks     | 5 | 9 | 10 | 15 | 20 | 25 |
| Frequency | 3 | 4 | 5  | 1  | 1  | 2  |

From the table above, the highest frequency is 5 and this corresponds to a mark of 10  
 $\therefore$  the mode is 10

### Example 2;

For a class of 30 students, the scores on a maths test out of 20 marks were as follows

8 10 14 4 6 12 10 10 16 18  
 10 8 4 6 14 18 16 14 14 14  
 6 8 10 10 4 6 12 14 14 4

Solution:

| Marks | Frequency |
|-------|-----------|
| 4     | 4         |
| 6     | 4         |
| 8     | 3         |
| 10    | 6         |
| 12    | 2         |
| 14    | 7         |
| 16    | 2         |
| 18    | 2         |

The highest frequency is 7;  $\therefore$  modal score = 14

### Explain meaning of dispersion and define range, variance and standard deviation for ungrouped data.

The variance is the arithmetic mean of the squares of the deviation of the observations from the true mean. It is also called the **mean squared deviation**.

The formula for variance is (a)  $\frac{\sum(x-\bar{x})^2}{n}$  for an ordinary distribution (ungrouped)

(b)  $\frac{\sum f(x-\bar{x})^2}{\sum f}$  for a frequency distribution table (grouped)

Standard deviation (S.D) is the square root of variance.

The formula for S.D are: (a)  $\sqrt{\frac{\sum (x-\bar{x})^2}{n}}$  and (b)  $\sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$

Example:

Find the variance and standard deviation of the set of numbers 2,5,6,3 and 4

Solution: Variance  $= \frac{\sum (x - \bar{x})^2}{n}$

But mean  $= 4$

| x | $x - \bar{x}$ | $(x - \bar{x})^2$           |
|---|---------------|-----------------------------|
| 2 | -2            | 4                           |
| 5 | 1             | 1                           |
| 6 | 2             | 4                           |
| 3 | -1            | 1                           |
| 4 | 0             | 0                           |
|   |               | $\sum (x - \bar{x})^2 = 10$ |

Variance  $= \frac{10}{5} = 2$

Standard deviation, S.D  $= \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$   
 $= \sqrt{2}$

$= 1.414$

Example 2:

Calculate the standard deviation of the distribution

|             |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|
| Age (years) | 10 | 12 | 13 | 14 | 15 | 16 |
| Frequency   | 18 | 4  | 6  | 12 | 6  | 4  |

Solution:

Reference to example 2 n page 3 and 4



$$\begin{aligned}
 \text{Standard Deviation} &= \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}} \\
 &= \sqrt{\frac{228.4}{50}} \\
 &= \sqrt{4.568} \\
 &= 2.14
 \end{aligned}$$

## Measures of Dispersion

The measure of dispersion (also called measure of variation) is concerned with the degree of spread of the numerical value of a distribution.

**Range:** This is the difference between the maximum and minimum values in the data.

Examples 1:

Find the range of the data 6, 6, 7, 9, 11, 13, 16, 21 and 32

Solution: The maximum item is 32

The minimum item is 6

$$\therefore \text{Range} = 32 - 6 = 26$$

Example 2:

Find the range of the distributions below 65, 62, 62, 61, 61, 60, 60, 59, 58, 52

$$\text{Solution: Range} = 65 - 52 = 13$$

Deviation from the mean:

If the mean of a distribution is subtracted from any value in the distribution, the result is called the DEVIATION of the value from the mean.

Consider the table below (set of examination marks)

|    |    |    |    |    |
|----|----|----|----|----|
| 65 | 62 | 62 | 61 | 61 |
| 60 | 60 | 59 | 58 | 52 |

$$\begin{aligned}\text{The mean} &= \frac{65 + 62 + 62 + 61 + 61 + 60 + 60 + 59 + 58 + 52}{10} \\ &= \frac{600}{10} \\ &= 60\end{aligned}$$

$$\begin{aligned}
 \text{Deviation from the mean} &= X - \bar{X} = 65 - 60 = +5 \\
 &= 62 - 60 = +2 \\
 &= 62 - 60 = +2 \\
 &= 61 - 60 = +1 \\
 &= 61 - 60 = +1 \\
 &= 60 - 60 = 0 \quad \text{e.t.c}
 \end{aligned}$$

The deviations of the scores from the mean are +5, +2, +2, +1, +1, 0, 0, -1, -2, -8. The sum of these deviations = 0